



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

DIFFERENTIAL AND INTEGRAL CALCULUS 1M — 104018 WINTER 2021

Midterm

November 20, 2021

Student ID Number:

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Instructions

1. **Duration: 180 minutes**
2. This exam has **6 questions**, for a total of **100** points.
3. Write your answers clearly, completely and concisely in the space provided.
4. **Quote** all results that you use. **No credit will be given to answers which are not justified.**
5. Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed.
6. During the exam, a dictionary will be available upon your request to the proctors.

Question 1 (25 points) Let $f : \mathbb{R} \longrightarrow \mathbb{R}$.

$$x \longmapsto e^{x^3} - 1$$

- (a) Show that f is one-to-one. **Solution:** Let $x, y \in \mathbb{R}$ and suppose that $f(x) = f(y)$. Thus

$$e^{x^3} - 1 = e^{y^3} - 1 \iff e^{x^3} = e^{y^3} \iff x^3 = y^3 \iff x = y.$$

$\uparrow$ $\uparrow$
 b/c $x \mapsto e^x$ is 1-1 b/c $x \mapsto x^3$ is 1-1

The above shows that f is one-to-one.

- (b) Compute $\text{im}(f) = \{f(a) \mid a \in \mathbb{R}\}$, the image (or range) of f .
Solution: We know that the cubic map, $\mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x^3$, is onto and that the image of $x \in \mathbb{R} \mapsto e^x$ is $(0, +\infty)$. Therefore, $\{e^{x^3} \mid x \in \mathbb{R}\} = \{e^y \mid y \in \mathbb{R}\} = (0, +\infty)$. It follows that $\{e^{x^3} - 1 \mid x \in \mathbb{R}\} = (-1, +\infty)$.

- (c) Determine $f^{-1}(x)$, for all $x \in \text{im}(f)$, where $f^{-1} : \text{im}(f) \longrightarrow \mathbb{R}$ is the inverse of f . **Solution:** Let $x > -1$. Then

$$\begin{aligned} f^{-1}(x) = y &\iff x = f(y) \iff x = e^{y^3} - 1 \\ &\iff y^3 = \ln(x + 1) \iff Y = \sqrt[3]{\ln(x + 1)}. \end{aligned}$$

So, $f^{-1}(x) = \sqrt[3]{\ln(x + 1)}$, for all $x > -1$.

- (d) Compute the tangent line to the graph of f at the point $(0, 0)$.
Solution: The function f is differentiable in $\mathbb{R}$ and $f'(x) = 3x^2 e^{x^3}$, for all $x \in \mathbb{R}$. Thus the slope of the tangent line to the graph of f at the point $(0, 0)$ is $f'(0) = 0$ and the equation of this tangent line is $y = 0$. (Note that this already implies that f^{-1} is not differentiable at 0.)
- (e) Compute the derivative of $f^{-1}(x)$ where it exists. You can use the fact (proved in Lecture 11) that $(\ln x)' = \frac{1}{x}$, for $x > 0$. **Solution:**

From $f^{-1}(x) = \sqrt[3]{\ln(x+1)}$, for all $x > -1$, the fact that $\ln x$ is differentiable for all $x > 0$ and $\sqrt[3]{x}$ is differentiable for all $x \neq 0$, we deduce that $f^{-1}(x)$ is differentiable for all $x > -1$ except possibly at the point x for which $\ln(x+1) = 0$, i.e., at $x = 0$. Indeed, it had already been observed above that f^{-1} is not differentiable at 0. For $x \in (-1, 0) \cup (0, +\infty)$ we have

$$(f^{-1})'(x) = \frac{1}{3}(\ln(x+1))^{-2/3} \frac{1}{x+1}.$$

Question 2 (20 points) Compute the derivatives of the following functions, explaining all the steps:

(a) $f(x) = \sqrt{x}(\sin^2 x + \pi e^x)$;

Solution: The domain of f is $[0, +\infty)$. The function $\sqrt{x}$ is differentiable only for $x > 0$, whereas the functions $\sin^2 x$ and πe^x are differentiable everywhere. Thus, f is differentiable on $(0, +\infty)$ and, using the differentiation rules, we obtain

$$f'(x) = \frac{1}{2\sqrt{x}}(\sin^2 x + \pi e^x) + \sqrt{x}(2 \sin x \cos x + \pi e^x), \quad \text{for } x > 0.$$

(b) $f(x) = 2^{\sin x} + \sqrt{\sin 3x}$.

Solution: The domain of f is $\{x \in \mathbb{R} \mid \sin 3x \geq 0\} = \{x \in \mathbb{R} \mid -\pi/6 + 2k\pi/3 \leq x \leq \pi/6 + 2k\pi/3, \text{ for some } k \in \mathbb{Z}\}$ and f is differentiable on $\{x \in \mathbb{R} \mid \sin 3x > 0\}$. For x in this set, using the differentiation rules, we obtain

$$f'(x) = (\ln 2)(\cos x)2^{\sin x} + \frac{3 \cos 3x}{2\sqrt{\sin 3x}}.$$

Question 3 (15 points) Compute the following limits, explaining all the steps:

(a) $\lim_{x \rightarrow 0^+} \left(\frac{\sin(2x)}{\sin x} \right)^{\frac{1}{x}};$

Solution: First, we see that

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{\sin(2x)}{\sin x} &= \lim_{x \rightarrow 0^+} 2 \frac{\sin(2x)}{2x} \frac{x}{\sin x} \underset{y=2x}{=} 2 \lim_{y \rightarrow 0^+} \frac{\sin y}{y} \lim_{x \rightarrow 0^+} \frac{1}{\frac{\sin x}{x}} \\ &= 2 \cdot 1 \cdot \frac{1}{1} = 2; \end{aligned}$$

and $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty.$

Therefore, as the limit of the base is strictly larger than 1, we conclude that $\lim_{x \rightarrow 0^+} \left(\frac{\sin(2x)}{\sin x} \right)^{\frac{1}{x}} = +\infty.$

(b) $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\cos \frac{1}{x}};$

Solution: Although the limit $\lim_{x \rightarrow 0^+} \cos \frac{1}{x}$ does not exist, $\cos \frac{1}{x}$ is bounded and thus so is $e^{\cos \frac{1}{x}}$. Specifically,

$$e^{-1} \leq e^{\cos \frac{1}{x}} \leq e, \quad \text{for all } x \neq 0.$$

We have that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$, so, as a consequence of the “Squeeze Theorem”, we deduce that $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\cos \frac{1}{x}} = 0.$

(c) $\lim_{x \rightarrow 0^+} (\sin x) \cos(\ln(1 + \frac{1}{x})).$

Solution: Although the limit $\lim_{x \rightarrow 0^+} \cos(\ln(1 + \frac{1}{x}))$ does not exist, $\cos(\ln(1 + \frac{1}{x}))$ is bounded. We have that $\lim_{x \rightarrow 0^+} \sin x = 0$, so, as a consequence of the “Squeeze Theorem”, we deduce that $\lim_{x \rightarrow 0^+} (\sin x) \cos(\ln(1 + \frac{1}{x})) = 0.$

Question 4 (10 points) Let $a \in \mathbb{R}$ be constant and

$$f(x) = \frac{\sin(ax)}{x} + \frac{\sqrt{ax^2 + 1} - 1}{x}.$$

Find all positive real numbers a for which $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow +\infty} f(x)$.

Solution: We have:

$$\lim_{x \rightarrow 0} \frac{\sin(ax)}{x} = a \lim_{x \rightarrow 0} \frac{\sin(ax)}{ax} = a;$$

$\uparrow$
 $a \neq 0$

$$\lim_{x \rightarrow 0} \frac{\sqrt{ax^2 + 1} - 1}{x} = \lim_{x \rightarrow 0} \frac{ax^2}{x(\sqrt{ax^2 + 1} + 1)} = \lim_{x \rightarrow 0} \frac{ax}{\sqrt{ax^2 + 1} + 1} = \frac{0}{2} = 0.$$

(Note that the later limit can also be seen and computed as

$$\left. \frac{d}{dx} \right|_{x=0} \sqrt{ax^2 + 1}.)$$

Thus, $\lim_{x \rightarrow 0} f(x) = a$.

We also have:

$$\lim_{x \rightarrow +\infty} \frac{\sin(ax)}{x} = 0;$$

$\uparrow$
 $\frac{1}{x} \rightarrow 0$ and $\sin(ax)$ is bounded

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{ax^2 + 1} - 1}{x} = \lim_{x \rightarrow +\infty} (\sqrt{a + 1/x^2} - 1/x) = \sqrt{a}.$$

$\uparrow$
 $x = |x| = \sqrt{x^2}$ for $x \geq 0$

Thus, $\lim_{x \rightarrow +\infty} f(x) = \sqrt{a}$ and we conclude that we must have $a = \sqrt{a}$. Squaring both sides and recalling that $a \neq 0$, we get the solution $a = 1$.

Question 5 (10 points) Let $f, g, h : I \longrightarrow \mathbb{R}$ be functions defined on an open interval I and $a \in I$. Suppose that:

- (i) $f(x) \leq g(x) \leq h(x)$, for all $x \in I$;
- (ii) $f(a) = h(a)$;
- (iii) $f'(a) = h'(a)$.

Prove that g is differentiable at a and that $g'(a) = f'(a) = h'(a)$.

Solution: Since $f(a) \leq g(a) \leq h(a)$ and $f(a) = h(a)$, we get $g(a) = f(a) = h(a)$. Thus,

$$f(x) - f(a) \leq g(x) - g(a) \leq h(x) - h(a), \quad \text{for all } x \in I.$$

So, for $x \in I$ with $x > a$, we have $x - a > 0$ and

$$\frac{f(x) - f(a)}{x - a} \leq \frac{g(x) - g(a)}{x - a} \leq \frac{h(x) - h(a)}{x - a}.$$

Moreover, $\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} = f'(a) = h'(a) = \lim_{x \rightarrow a^+} \frac{h(x) - h(a)}{x - a}$.

Then, it follows by the “Squeeze Theorem” that

$$\lim_{x \rightarrow a^+} \frac{g(x) - g(a)}{x - a} = f'(a) = h'(a).$$

Similarly, for $x \in I$ with $x < a$, we have $x - a < 0$ and

$$\frac{h(x) - h(a)}{x - a} \leq \frac{g(x) - g(a)}{x - a} \leq \frac{f(x) - f(a)}{x - a}.$$

Since $\lim_{x \rightarrow a^-} \frac{h(x) - h(a)}{x - a} = h'(a) = f'(a) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$, again the

“Squeeze Theorem” implies that

$$\lim_{x \rightarrow a^-} \frac{g(x) - g(a)}{x - a} = f'(a) = h'(a).$$

In conclusion, $\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}$ exists and

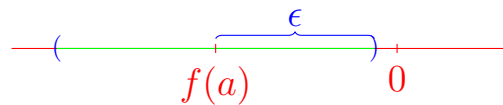
$$g'(a) = \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} = f'(a) = h'(a).$$

Question 6 (20 points) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $a \in \mathbb{R}$.

- (a) Show that if f is continuous at a and $f(a) < 0$, then there exists $\delta > 0$ such that $f(x) < 0$ for all $x \in (a - \delta, a + \delta)$.

Solution: Since f is continuous at a , we have $\lim_{x \rightarrow a} f(x) = f(a)$.
From the definition of limit, this means that

$$(\forall \epsilon > 0) (\exists \delta > 0) (\forall x \in \text{Dom}(f)) \quad 0 < |x - a| < \delta \implies |f(x) - f(a)| < \epsilon.$$



Consider the illustration above as an aid for the following argument: if $|f(x) - f(a)| < \epsilon$, then

$$f(x) = (f(x) - f(a)) + f(a) \leq |f(x) - f(a)| + f(a) < \epsilon + f(a).$$

If we choose any $0 < \epsilon \leq -f(a)$, e.g. $\epsilon = -f(a)$ (this is possible because $f(a) < 0$), then the above guarantees the existence of $\delta > 0$ such that, for all $a - \delta < x < a + \delta$ with $x \neq a$, we have $f(x) < \epsilon + f(a) = 0$. In case $x = a$, then we also have $f(a) < 0$, so we have

$$f(x) < 0 \quad \text{for all } x \in (a - \delta, a + \delta).$$

- (b) Show, by providing an example, that the previous statement is false if f is not continuous at a .

Solution: Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x \neq a; \\ -1 & \text{if } x = a. \end{cases}$$

Then $f(a) < 0$ but any open interval I which contains a (such as $(a - \delta, a + \delta)$, for some $\delta > 0$) also contains some other point $x \neq a$, and thus $f(x) \not\leq 0$.